

WEEK 4 – POWER BI EDA

Exploratory Data Analysis and Simple Moving Average Dataset: Shopify Stock Data

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Introduction

This week focuses on performing **Exploratory Data Analysis (EDA)** on Shopify stock-market data using Power BI. The analysis aims to understand trading activity and identify trends in stock volume over time. Simple Moving Average (SMA) calculations will be created using DAX for **20-day and 50-day periods**. These moving averages will help smooth daily fluctuations and provide a clearer view of trading-volume trends. The existing Date Dimension and data model will be used to perform time-based analysis and create interactive visualizations in Power BI.

Figure 1: New Week 4 EDA report page created for Shopify stock analysis

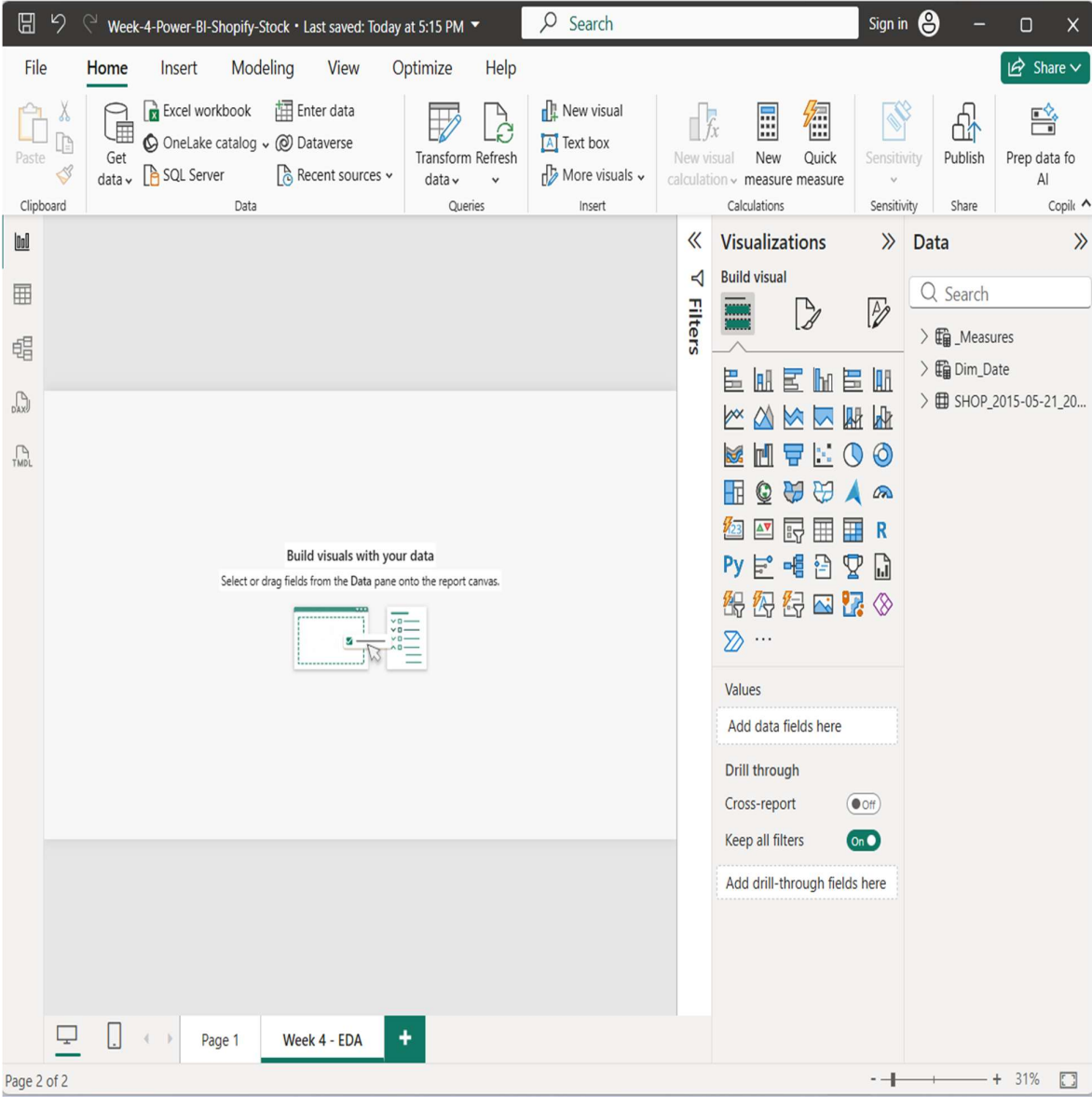


Figure 2: Existing Shopify stock data model and active Dim_Date relationship reused for Week 4 EDA

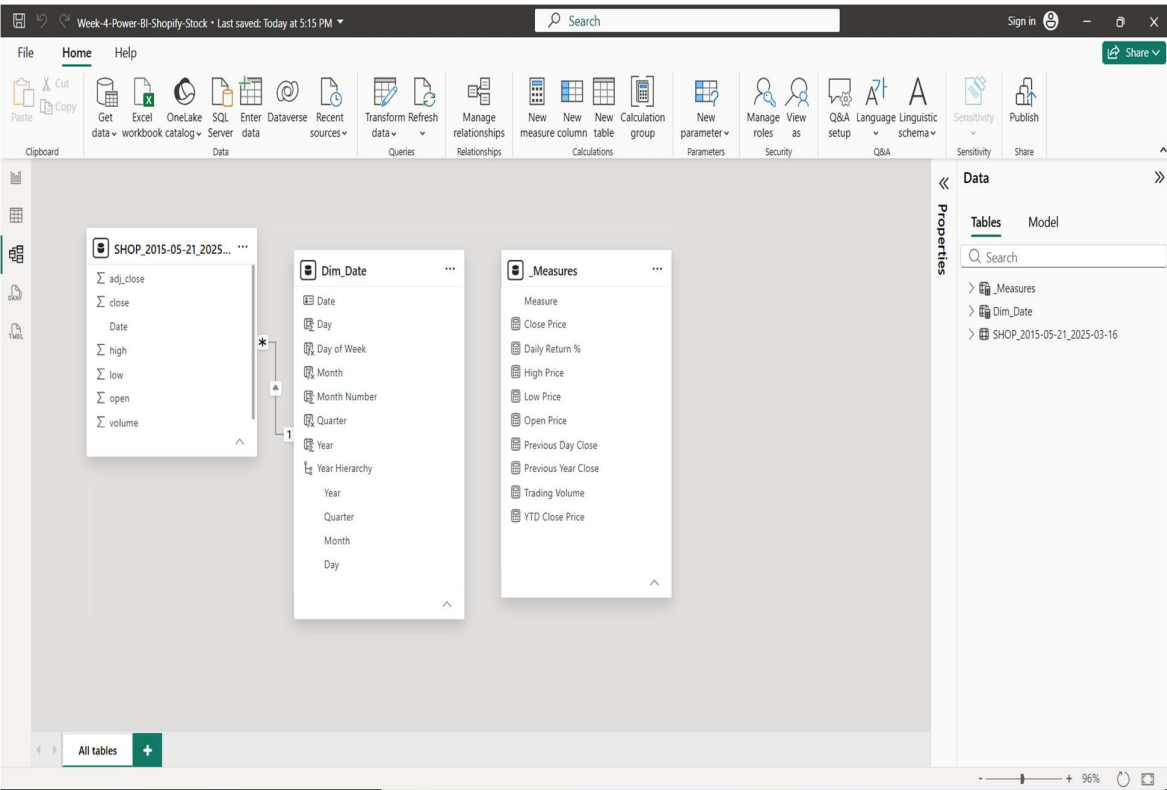


Figure 3: 20-Day Simple Moving Average of Shopify trading volume calculated using DAX

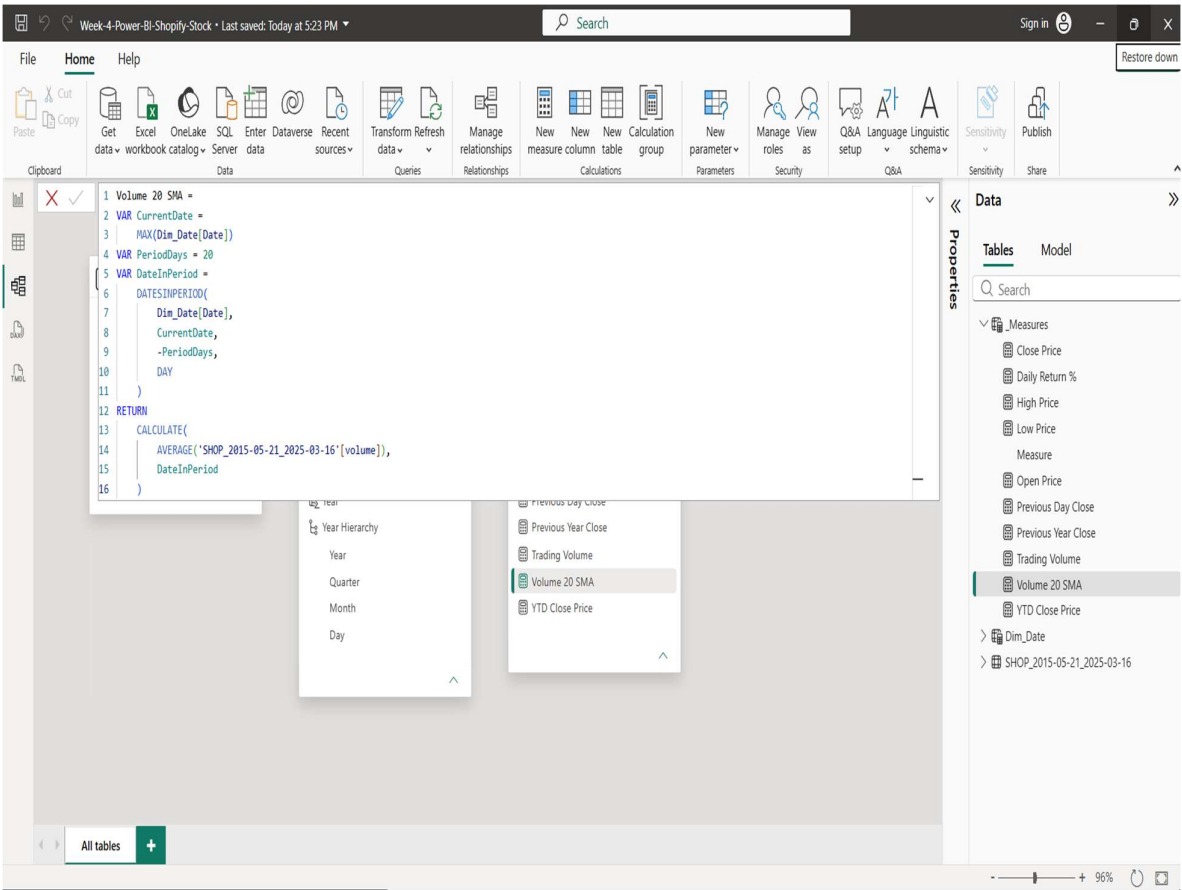


Figure 4: 50-Day Simple Moving Average of Shopify trading volume calculated using DAX

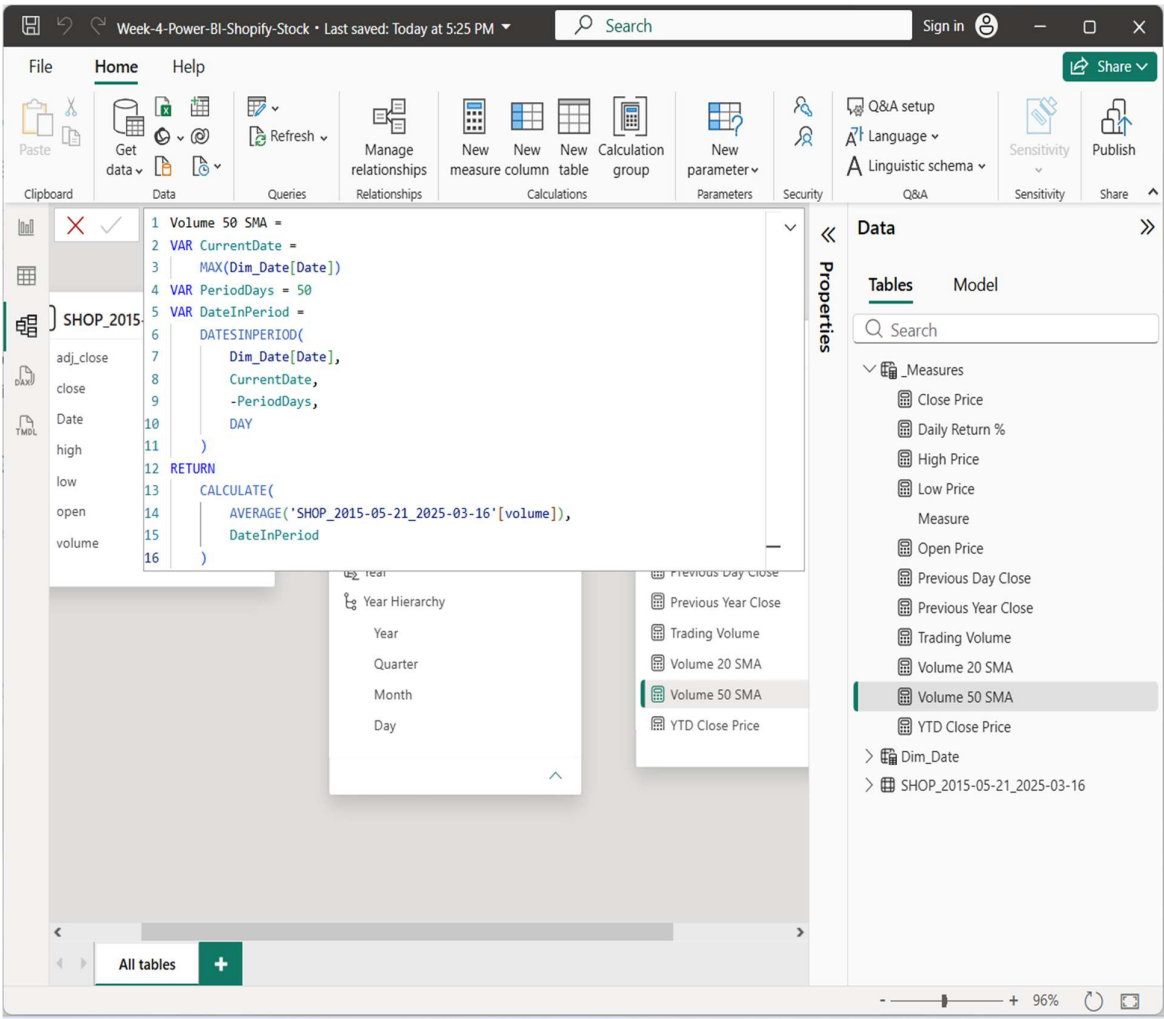


Figure 5: Shopify trading volume with 20-Day and 50-Day Simple Moving Averages

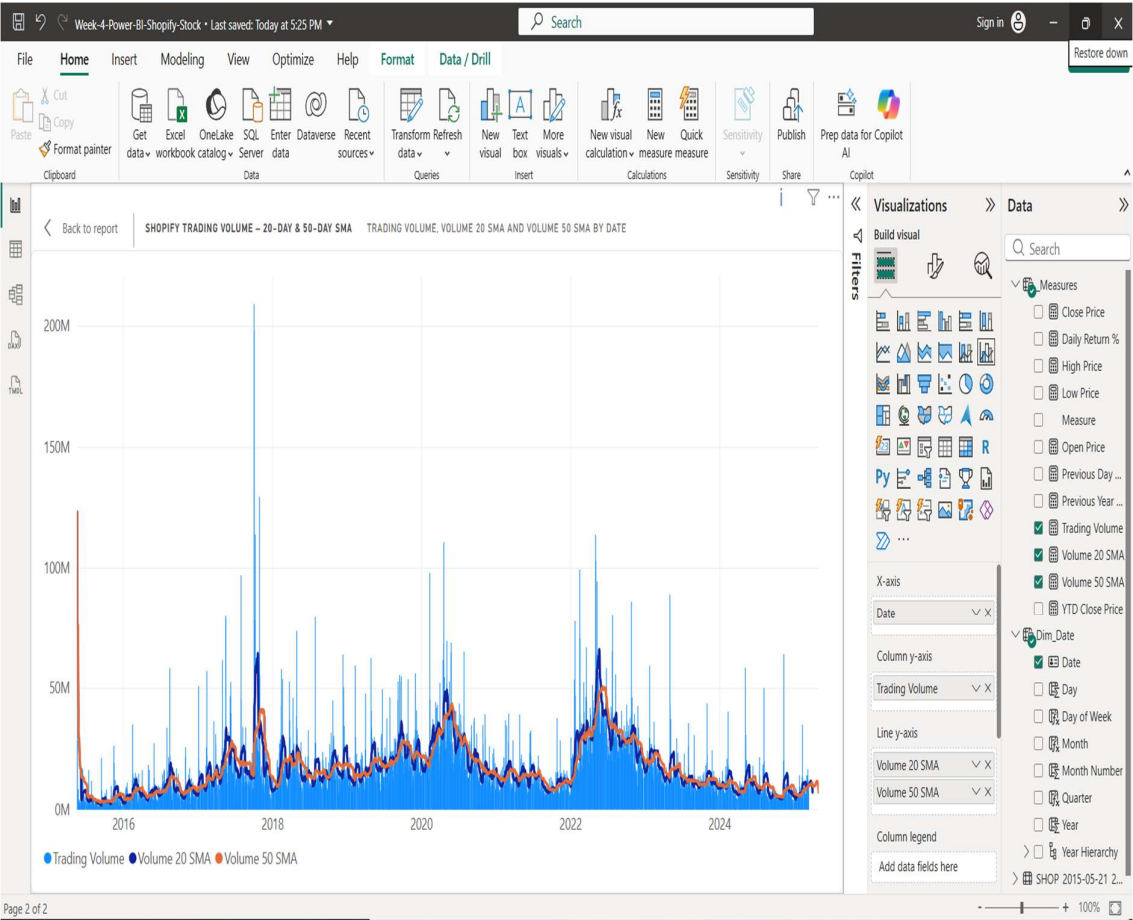


Figure 6: Comparison of 20-Day and 50-Day Simple Moving Average trends

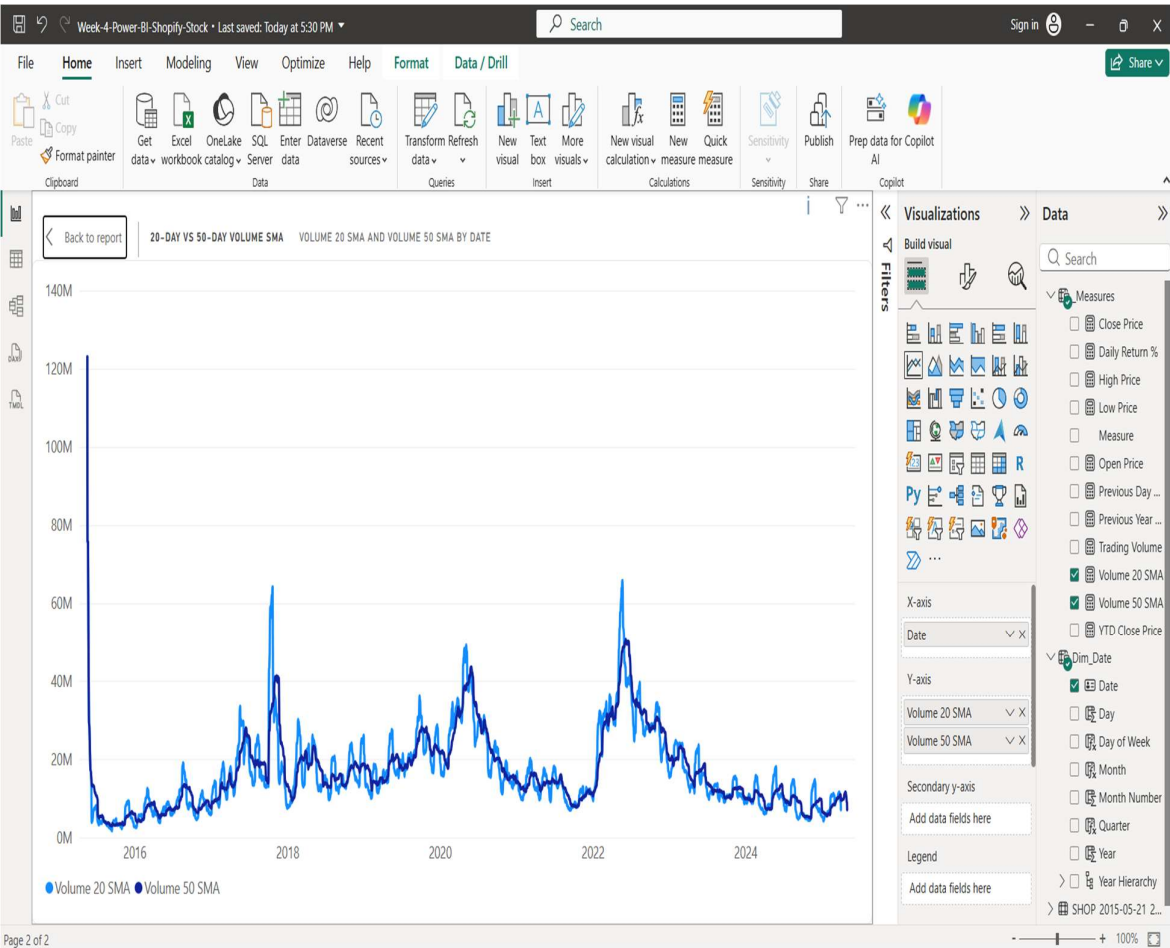


Figure 7: Interactive date slicer used to filter Shopify volume and moving-average analysis

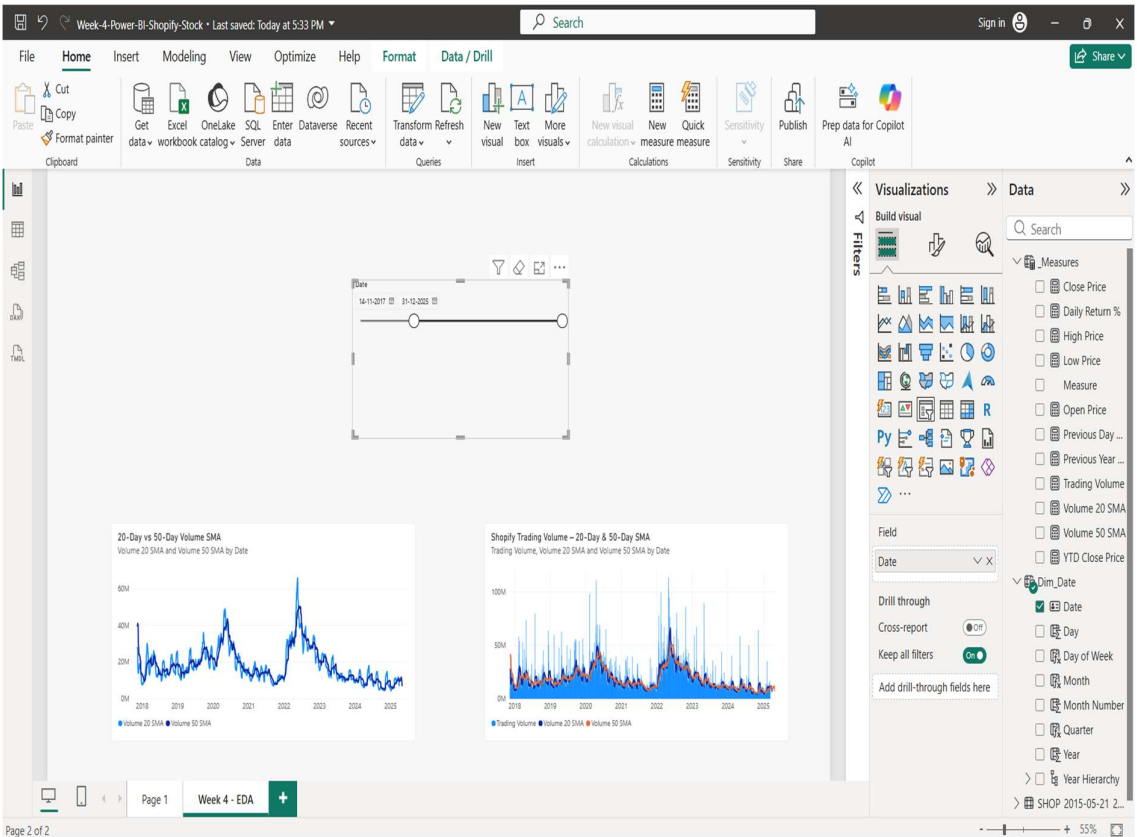


Figure 8: Trading volume KPI displayed alongside Shopify moving-average analysis

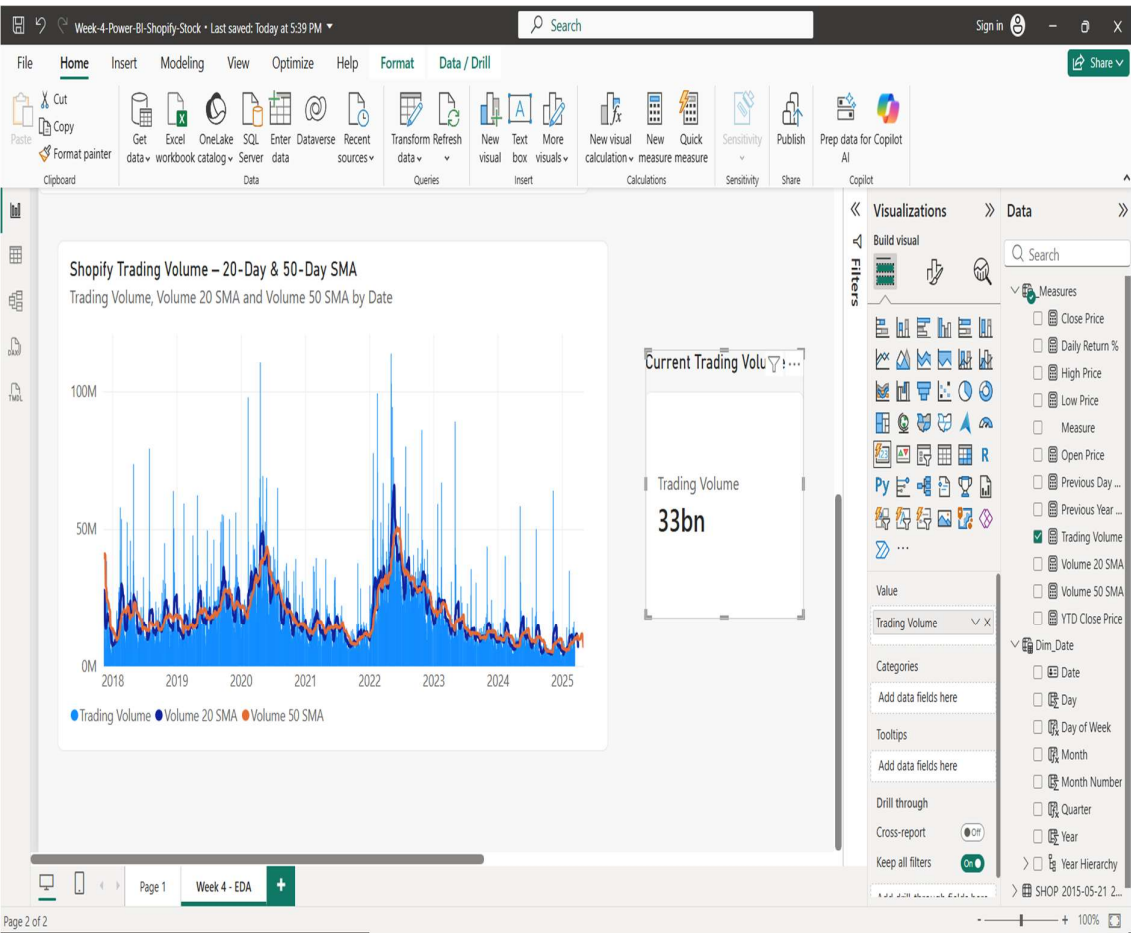


Figure 9: Week 4 Shopify Stock Exploratory Data Analysis dashboard with 20-Day and 50-Day SMA

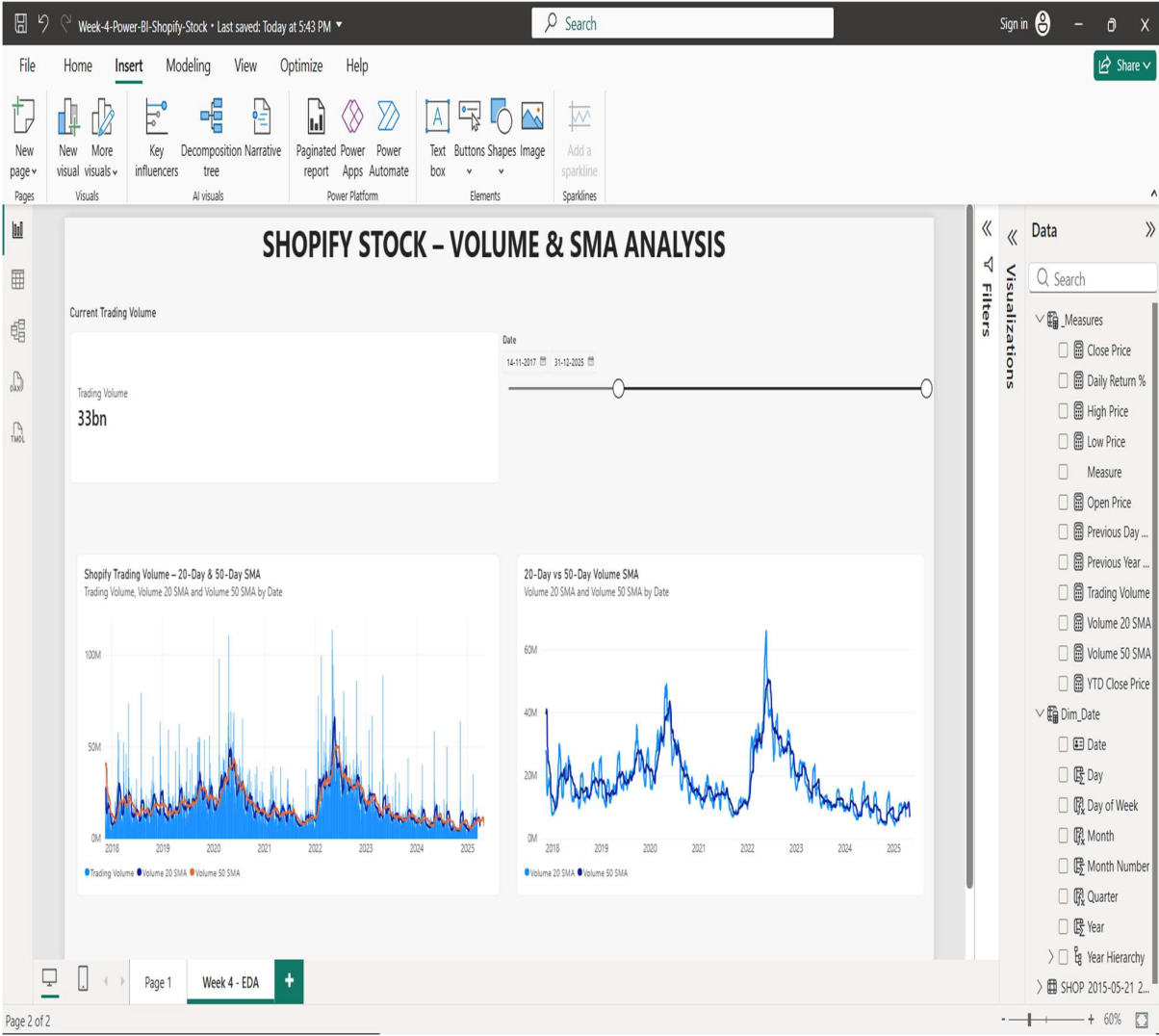
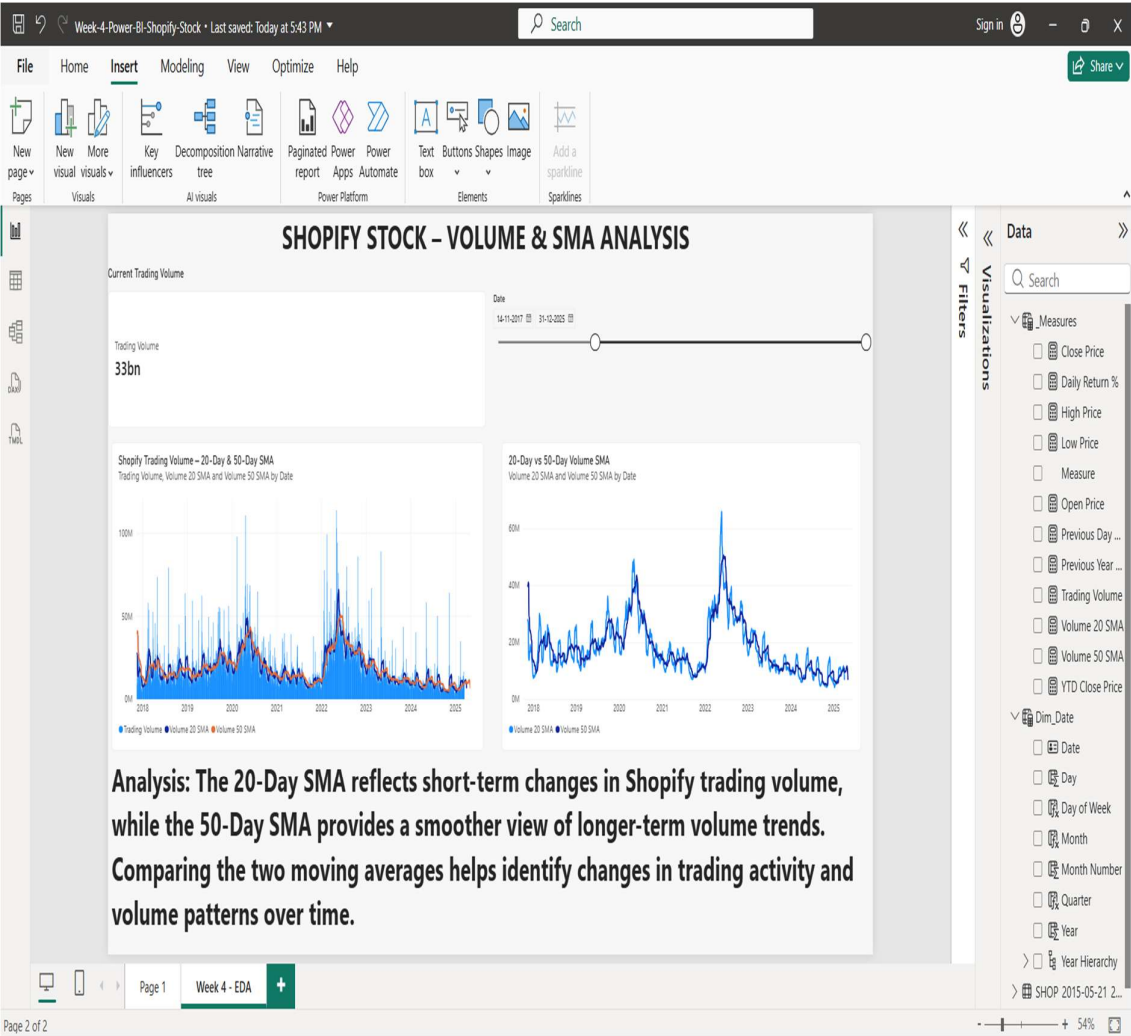


Figure 10: Interpretation of 20-Day and 50-Day Simple Moving Average trends



Conclusion

The Shopify stock dataset was analyzed using Power BI to perform Exploratory Data Analysis (EDA). The trading volume data was evaluated using **20-Day and 50-Day Simple Moving Averages (SMA)** created with DAX. The moving averages helped smooth daily fluctuations and provided a clearer view of short-term and longer-term trading-volume trends. Interactive visualizations, a date slicer, and a trading-volume KPI were created to explore the data over different time periods. The comparison between the 20-Day and 50-Day SMA provides a useful way to observe changes in Shopify's trading activity over time. Overall, the Week 4 analysis demonstrates the use of **DAX, time-based filtering, moving averages, and interactive Power BI visualizations** for financial Exploratory Data Analysis.